

Assignment # 1

Deadline- 15 April 2026

All questions should be handwritten, except for the laboratory assignment (Q-6). Scan all responses and compile them into a single PDF file, then submit via email to draj28@yahoo.com

1. Explain the core functions of a Geographic Information System (GIS). Describe how GPS and Remote Sensing technologies work for spatial data acquisition and explain how the collected data is integrated into a GIS environment?
2. Define map scale and spatial resolution with appropriate illustrations and discuss their importance in GIS analysis. If the scale on a map is 1:25,000, (i) find the distance in meters between two places represented by a distance of 1mm on the map and (ii) find the distance between two places on the map that are 50 kilometers apart?
3. Describe point-in-polygon and line-in-polygon overlay operations with suitable diagrams and attribute tables. Explain why a line-in-polygon overlay operation typically produces more features than the input line layer. Provide real-world GIS application examples.
4. Differentiate between raster and vector data models. Explain different raster image overlay operations such as Boolean overlay, arithmetic overlay, weighted overlay, and conditional overlay.
5. What are the different steps in remote sensing? Describe with illustrations. Differentiate between active and passive sensor?
6. Using QGIS, create a small GIS database for a selected study area (e.g., Tribhuwan University, Kirtipur Municipality).
 - a. Acquire spatial data using at least three sources:
 - GPS/Google Earth point collection (digitized points)
 - Remote sensing image (Sentinel/Landsat/Google satellite)
 - Existing vector data (OpenStreetMap or shapefile)
 - b. Import all datasets into QGIS
 - c. Create vector layers for:
 - Points (important locations)
 - Line (road network)
 - Polygon (study boundary or land use)
 - d. Create attribute tables and add at least 5 fields
 - e. Prepare a map layout showing all layers